

NASA SPACE APPS CHALLENGE 2026 · TEAM BRIEFING

EarthScope

Earth System Trend Detective — the complete project report, A to Z

Live app
earthscope-climate.vercel.app

Focus
Dhaka & Bangladesh, works for any place on Earth

Challenge
NASA Space Apps 2026 · Be An Earth System Trend Detective! · 4 Oct 2026

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How to read this report

Everyone on the team should read sections **1–2** (what we built and how it answers the challenge), **7** (our findings) and **19** (how to present and defend it). Sections **3–6** explain the app and the science; developers continue with **9–15**. Section **17** lists what EarthScope does *not* claim — know it before talking to judges.

> **One-sentence pitch.** EarthScope is an Earth-system trend detective: for any NASA variable it maps *what* is changing, *where*, *how much* and whether it is *significant* — and catches the data artefacts that fool naive analyses — in English and বাংলা, built for Dhaka.

1 Executive summary — everything we built

EarthScope is a live web app (earthscope-climate.vercel.app) that turns NASA Earth-observation and model data into answers anyone can check: **what is changing, where, how much, and is it significant?** It is built for Dhaka and Bangladesh, works for any place on Earth, and runs fully in English and বাংলা.

13
NASA variables trend-mapped (10 MERRA-2 model variables + 3 MODIS satellite products)

~700
grid cells or 25,600 satellite pixels tested at once, with false-discovery-rate control

51
automated tests, including real-data regression tests

8 pages
Explorer, Trend Lab, Disasters, Atlas, Satellite, Future, Compare, Reports

950+
interface strings in Bangla — 0 missing

0 keys
needed — every source is open; free hosting

The eight parts of EarthScope

Part	In one line
Trend Lab CHALLENGE CORE	Maps significant trends for 10 NASA POWER variables in 5 seasons and for MODIS surface temperature (day/night) and greenness, with Mann–Kendall + Sen's slope + FDR, per-cell break testing and a live "one driver, opposite trends" case study.
Earth explorer	3D globe with daily real clouds; 45-year record of any place with the trend detective, data-quality check and ERA5 cross-check.
Live disasters	Storms, floods, quakes, fires right now; "Am I safe?", shelters outside danger zones, heat and air-quality warnings, alerts, offline safety pack.
Climate atlas	World → country → district choropleths 1981–2025 with play-through and rankings.
Satellite comparison	Measured land-cover, vegetation and surface-heat change, time-lapse 2001–2024, swipe, 30 m imagery.
Future scenarios	What-if sliders and IPCC AR6 sea-level projections next to ground height.
Compare & Reports	Two places side by side; 8-page bilingual reports (citizen / student / planner) with PDF, CSV and QR.
Ask Earth	AI guide grounded only in EarthScope's data, answering in English or Bangla, with an offline fallback.

Three findings to remember

1. **Many "trends" in raw data near Bangladesh are partly artefacts.** 224 of 693 NASA POWER grid cells around Bangladesh contain a non-climatic step (most often 2008). Without break testing, 21% of the area would look significantly *cooling*; with it, 8% — and 36% significantly warming.
2. **Dhaka is warming, but raw NASA POWER hides it.** POWER shows a false 1997 drop; the independent ERA5 dataset confirms it is an artefact and shows +0.21 °C per decade (1981–2025).
3. **One driver, opposite trends.** Around Dhaka, every March since 2001, irrigated fields cool by day (−0.58 °C/decade) yet warm by night (+0.41 °C/decade) and green up strongly (79% of pixels significant), while the city–countryside contrast grows by +0.55 °C/decade.

2 How we answer the challenge, line by line

> **The challenge.** "Find and examine variables measured by NASA missions or produced by NASA models, visualize how they change over time, and determine what is changing, where it is changing, how much it is changing, and if these changes are what scientists call 'significant'. … A variable can trend one way in one region and the opposite way in another, even when the same process drives both."

Challenge asks	EarthScope's answer	Where
Variables from NASA missions or NASA models	Model: NASA POWER / MERRA-2 (air temperature, max and min temperature, rainfall, relative and specific humidity, root-zone soil wetness, wind, sunshine, cloud). Missions: MODIS Terra (day and night land-surface temperature, NDVI, land cover), VIIRS, Landsat/HLS, GPM, GISTEMP, NSIDC.	Trend Lab, Explorer, Satellite
Visualize how they change over time	Trend maps, time series with Sen trend lines for any cell or pixel, climate stripes, time-lapse 2001–2024, globe time machine, atlas animation.	All pages
What is changing	Choose any of 13 variables and 5 seasons (whole year, winter, pre-monsoon, monsoon, post-monsoon) or any month for satellite products.	Trend Lab
Where	Every grid cell (0.5° × 0.625°) or 1–2 km pixel is mapped: red rising, blue falling, dots where significant; the land-cover breakdown says *on what kind of land*.	Trend Lab map
How much	Sen's slope per decade (robust median of all pairwise slopes), with % of area rising/falling and per-class means.	Trend Lab panel
Is it significant?	Mann–Kendall test with trend-free pre-whitening (autocorrelation) and the Benjamini–Hochberg false-discovery rate across the map, plus a break test so data artefacts are not called trends. Single places also get a 95% interval and an autocorrelation-corrected t-test.	Trend Lab, Explorer
Opposite trends in different regions, same process	The map flags when one variable rises in part of a region and falls in another. The live case study shows irrigation cooling fields by day and warming them by night, and city vs countryside diverging — one driver, opposite signs.	Trend Lab case study
Earth's interconnected system	The case study links three variables (daytime heat, night-time heat, vegetation) to one physical mechanism (evaporation from irrigated fields), with land cover explaining where.	Trend Lab

Why this can win

Scientific validity

Standard climate-science methods (Mann–Kendall, Sen, pre-whitening, FDR), break detection that catches artefacts most teams will miss, an independent ERA5 cross-check, and 51 tests.

Relevance and impact

Built for Bangladesh — heat, irrigation, floods, air, sea level — in Bangla, with live safety tools a citizen can use today.

Creativity

"Turn the break test off" shows judges, live, how a naive analysis would mislead. The opposite-trends case study is computed live from NASA pixels in about 20 seconds.

Presentation

A polished, fast, installable app; every number shows its source and method; bilingual reports for decision-makers.

3 The project in one page

EarthScope is a free web app (it also installs on a phone) that answers four questions for any place: **How has the climate here changed? What is dangerous right now? How has the land changed? What could the future look like?**

45 years
of NASA climate data per place (1981–2025), break-tested

7 pages
Explorer, Live disasters, Atlas, Satellite, Future, Compare, Reports

20+ sources
NASA, ECMWF, Copernicus, IPCC, USGS, GDACS, OpenStreetMap…

2 languages
English and বাংলা — 830 interface strings, 0 missing

51 tests
automated checks of the science code, all passing

0 keys
needed to run — every data source is open

Who it is for

Citizens
"Is Dhaka really getting hotter? Is a cyclone coming? Where is the nearest shelter? Is the air safe today?"

Students
Charts explained in plain language, a glossary, real NASA imagery and honest statistics to learn from.

Planners & NGOs
Downloadable 8-page reports with full statistics, CSV data, land-cover and heat change, risk profile with evidence.

What makes it different

- **It refuses to mislead.** NASA POWER data for Bangladesh contain an artificial 1 °C drop in 1997. A naive app would tell Dhaka it is *cooling*. EarthScope detects the fault, excludes it, and confirms it against an independent dataset (ERA5).
- **Measured, not guessed.** Satellite maps are decoded pixel-by-pixel back to NASA's real values (land-cover class, vegetation index, surface temperature in 0.6 K steps).
- **Useful today, not only for 2050.** Live cyclones, floods, earthquakes and fires, heat and air-quality warnings, shelters outside danger zones, alerts for saved places and an offline safety pack.
- **Built for Bangladesh.** Full Bangla interface, Bangla AI answers, BMD heatwave classes, FFWC/DDM links, 999 and 1090 emergency numbers, delta-subsidence notes.

4 The problem and our answer

The problem

Bangladesh is among the countries most exposed to climate change: heat waves, monsoon floods, cyclones and storm surges, sea-level rise on a sinking delta, and some of the worst air in the world. The data to understand these risks exist — NASA and other agencies publish them openly — but they are scattered across dozens of portals, in technical formats, in English, and often with traps (data breaks, colour-coded images, uncertain trends) that lead non-experts to wrong conclusions.

Our answer

Collect
20+ open sources, fetched live by our server and cached sensibly.

→

Check
Break tests, ERA5 cross-check, significance tests, exact pixel decoding.

→

Explain
Plain-language story, charts, maps and AI answers — in English or বাংলা.

→

Act
Safety checks, shelters, alerts, offline pack, reports to share with decision-makers.

Challenge fit

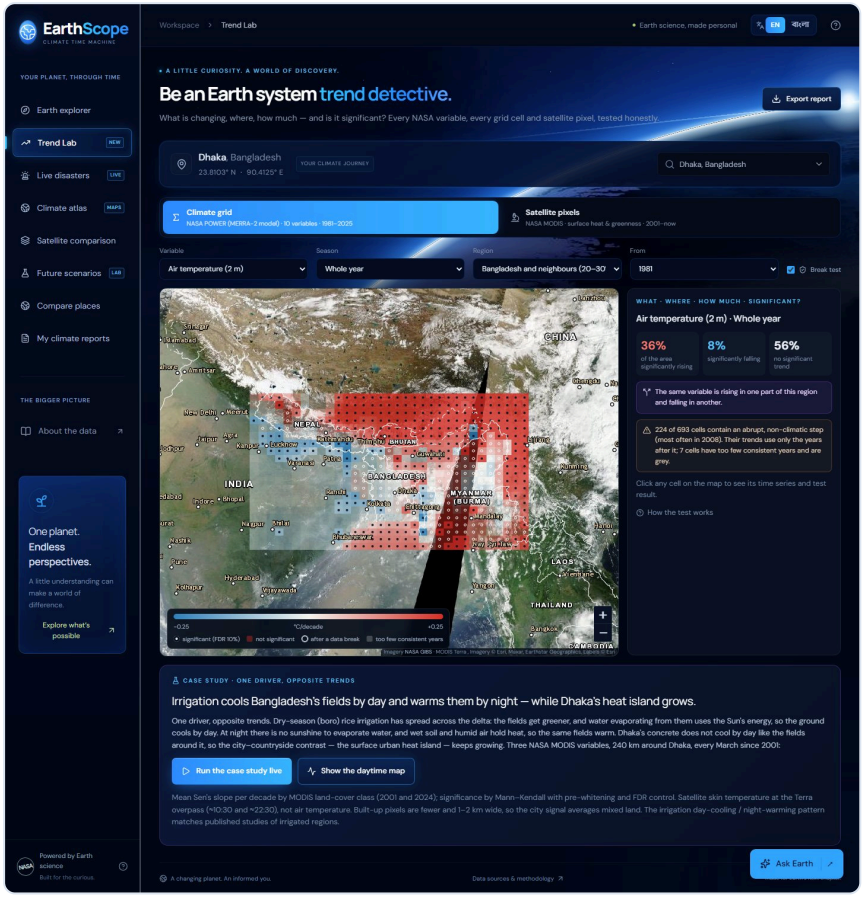
EarthScope uses NASA Earth-science data as its core (NASA POWER, GIBS imagery from MODIS, VIIRS, Landsat/HLS, GPM, EONET events, GISTEMP, SEDAC population, LANCE flood product) and adds partner data where NASA has no product for the job (ECMWF ERA5 for validation, Copernicus CAMS for air quality, IPCC AR6 for sea level, OpenStreetMap for shelters).

> **Design principles we kept throughout.** (1) Every number shows its source. (2) Never call a trend real unless it is statistically significant. (3) Never show a scenario as a forecast. (4) Never present EarthScope as an official warning service — always point to BMD/DDM and 999. (5) It must work in Bangla and on a cheap phone.

5 Tour of the app, page by page

Every page has its own web address, so a link can be shared directly. Heavy parts (3D globe, maps, report) load only when you open them.

5.1 Trend Lab `/trends` NEW



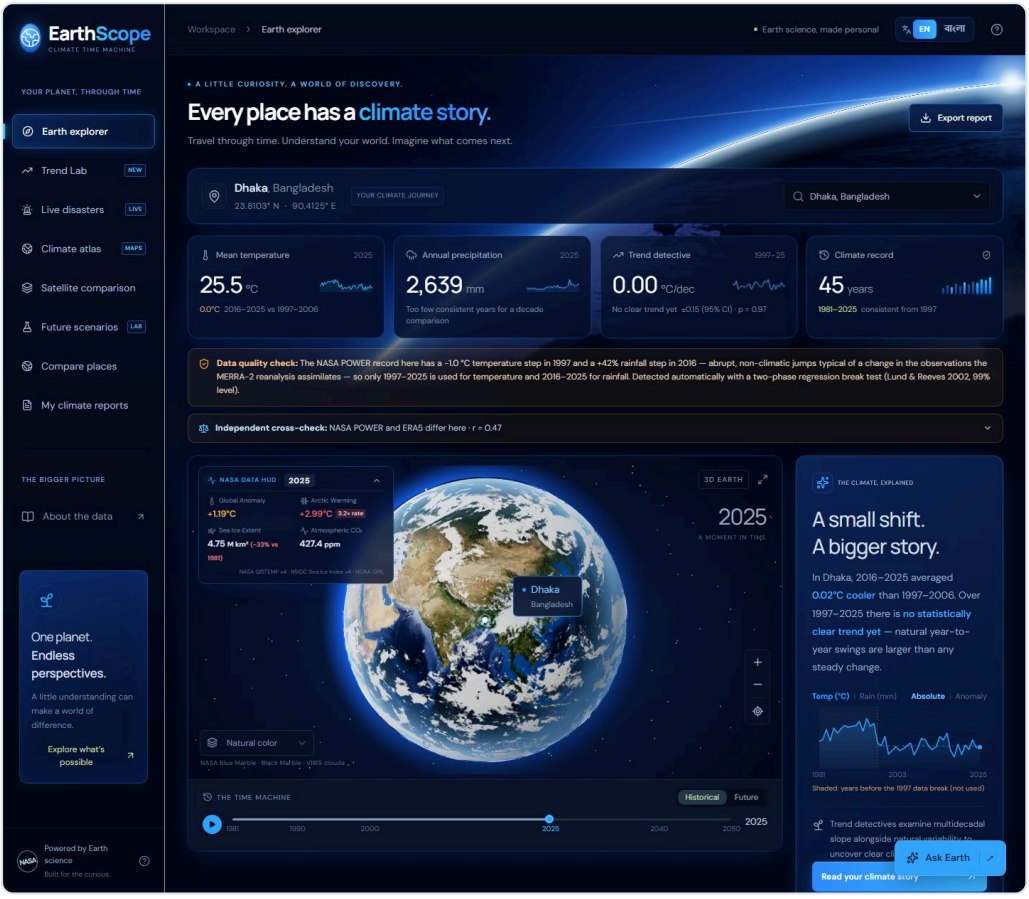
Trend Lab, live: significant air-temperature trends around Bangladesh (1981–2025, break-tested). Red = warming, blue = cooling, dots = significant after FDR control, rings = trend measured after a data break.

- **Climate grid** engine: NASA POWER (MERRA-2) — 10 variables, 5 seasons, three regions (Bangladesh and neighbours, South Asia, or a 10° square around any place), start year 1981/1991/2001.
- **Satellite pixels** engine: MODIS daytime and night-time surface temperature and NDVI, any month, 45/90/240 km around any place, every year 2001–now; results broken down by MODIS land cover (built-up since 2001, newly built, cropland, other).
- Click any cell or pixel: its time series, Sen trend line, p-value and any data break.
- **Break test switch**: turn it off to see what a naive analysis would claim.
- **Case study**: one click runs the three-variable "one driver, opposite trends" analysis live.

5.2 Earth explorer `/`

- **Realistic 3D globe** (three.js): NASA Blue Marble by day, VIIRS city lights by night, a moving day/night line, atmosphere glow, stars — and **real cloud cover from two days ago**, rebuilt every day from NASA satellites (§4.8). Click any place to explore it.
- **Place search** for any town or city worldwide (Open-Meteo / GeoNames).

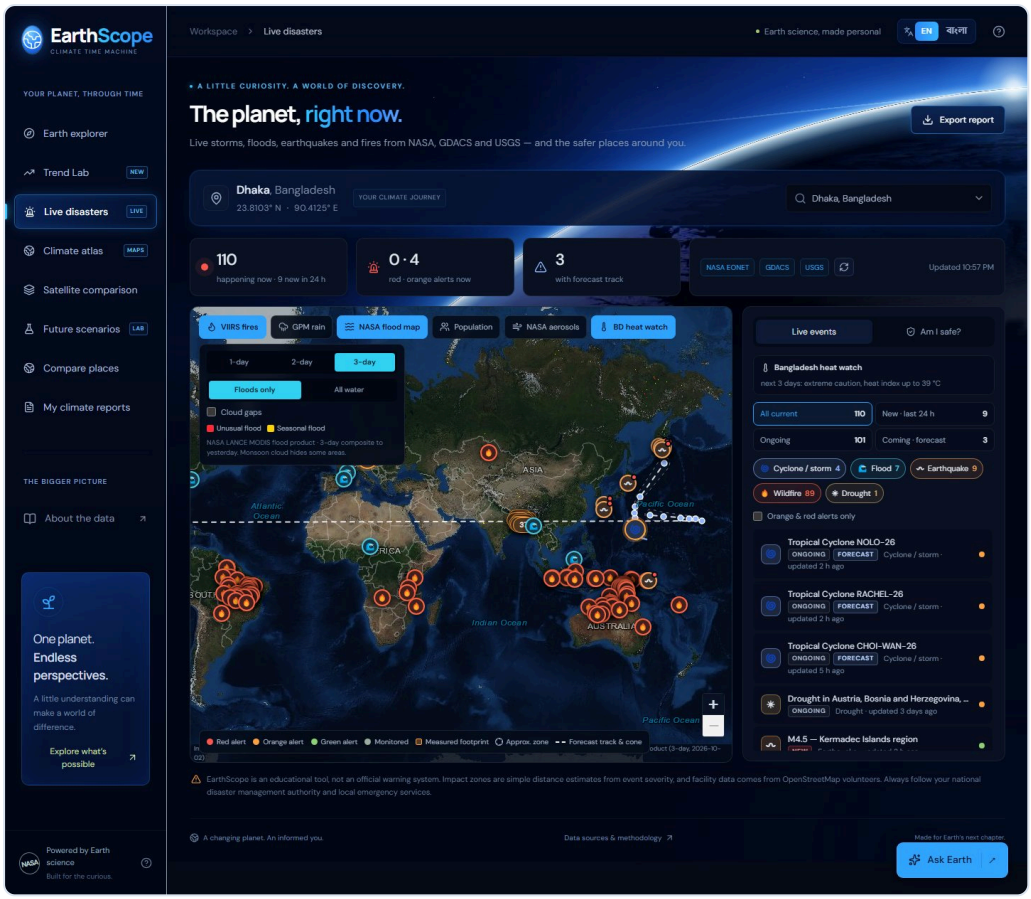
- **Metric cards**: mean temperature, rainfall, the **Trend detective** (°C per decade with 95% interval and p-value) and the length of the consistent record.
- **Data-quality note** when a data break is found, and the **ERA5 cross-check** card (§4.2).
- **Time machine**: a year slider 1981–2050 with play button; globe layers for temperature, rainfall, vegetation and flood exposure; a NASA data panel (global temperature anomaly, Arctic warming, sea-ice extent, CO_2).
- **Your climate story**: a plain-language narrative of the place, written natively in Bangla when Bangla is selected.



Earth explorer for Dhaka: metric cards, data-quality note (1997 and 2016 breaks), ERA5 cross-check, 3D globe with real clouds and the climate story.

5.3 Live disasters `/disasters`

- Map of what is happening **now**: storms, floods, earthquakes, wildfires, volcanoes, drought, dust, from NASA EONET, GDACS and USGS (refreshed every 5 minutes). Official cyclone forecast tracks with times, uncertainty cone and wind zones.
- NASA layers: VIIRS fires, GPM rain, the LANCE MODIS **flood map** (recoloured so unusual floods stand out), SEDAC population, **MODIS aerosols** (smoke, dust, haze) and a Bangladesh heat-watch layer.
- **Am I safe?** — checks your GPS location, a typed place or a map click against every hazard zone; gives protective actions for that hazard, the shortest way out of the zone, and the nearest shelters, hospitals and assembly points *outside* the zones with walking routes. Also shows a 7-day **heat outlook** (heat index, BMD classes) and **air quality** (CAMS forecast, US AQI, WHO PM2.5 comparison, health advice).
- **Event impact**: estimated people exposed (SEDAC population) and storm-surge heights for cyclones.
- **My places**: save Home/Work/School/Family; EarthScope alerts you (notifications) if a hazard zone, a cyclone forecast or dangerous heat reaches them, and keeps an **offline safety pack** that can be shared by SMS.

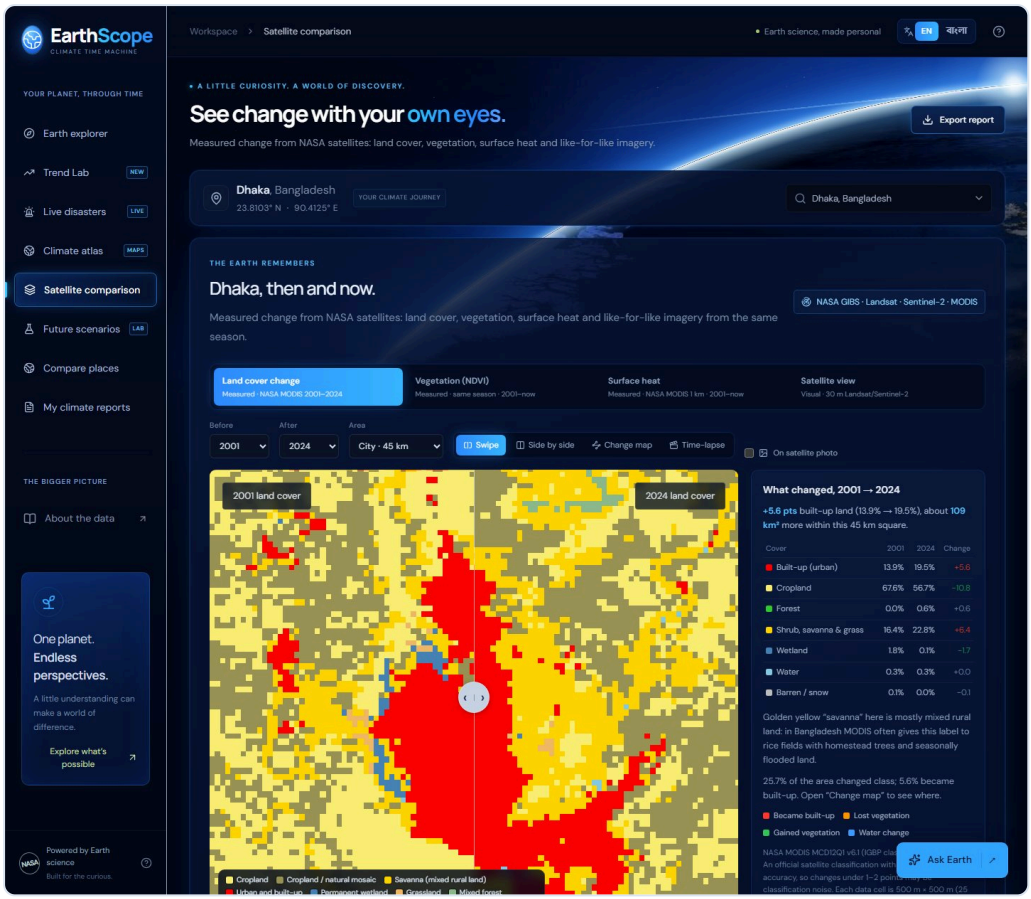


Live disasters: 110 current events from NASA EONET, GDACS and USGS, cyclone forecast tracks, NASA flood map, Bangladesh heat watch.

5.4 Climate atlas `/atlas`

A choropleth map that zooms World → country → states/divisions → districts. Shows temperature change, mean temperature, rainfall change or annual rainfall for every year 1981–2025, with play-through animation, climate stripes, a fastest-warming ranking and NASA imagery of that year. Each region is break-tested separately.

5.5 Satellite comparison `/satellite`



Satellite comparison: MODIS land cover 2001 vs 2024 around Dhaka — +5.6 points built-up (≈109 km²), with map key and swipe.

Compares two years for the same square (45, 90 or 240 km). Four modes:

Mode	What it measures	NASA product
Land cover change	Share of built-up, cropland, forest, water, etc. in each year; km² of new built-up land; change map. **Time-lapse** plays every year 2001–2024 with a built-up growth chart. On-map colour key; optional overlay on a present-day satellite photo.	MODIS MCD12Q1 (IGBP), 500 m, yearly
Vegetation (NDVI)	Greenness in the same season of both years (3 composites averaged), dense-vegetation share, greener/browner areas.	MODIS MOD13Q1, 250 m, 16-day
Surface heat	Daytime ground/roof temperature, area mean, hot-spot intensity (urban heat island), and where land warmed *more than its surroundings*.	MODIS MOD11A2 LST, 1 km, 8-day
Satellite view	Like-for-like 30 m photos: automatic search for the clearest pass in the same month of both years; haze removal.	HLS Landsat 8/9 + Sentinel-2; MODIS 7-2-1

Views: **Swipe** (drag the handle), **Side by side**, **Change map** and (land cover) **Time-lapse**.

5.6 Future scenarios `/future`

- Transparent “what-if” sliders: extra warming (0–4 °C) and rainfall change (±40%) by 2050, applied to the latest observation. Clearly labelled as an educational sensitivity test, **not** a climate-model projection.
- **Sea-level panel**: IPCC AR6 global sea-level projections for low, intermediate and very high emissions (2050 and 2100, with likely ranges), next to the place's **ground height** (Copernicus 90 m elevation model) and a verdict for its elevation band. In the Bengal delta it adds the land-subsidence warning.

5.7 Compare places `/compare`

Two climate records side by side (same NASA variables, same years), each break-tested, with a statistical verdict on whether one is really changing faster than the other (overlapping 95% intervals).

5.8 Climate reports `/reports`

An 8-page A4 assessment for any place: cover and key findings in IPCC likelihood language; climate record charts; climate extremes from daily data (hot days, warm nights, humid-heat days, heavy rain, dry spells); satellite evidence; a risk profile where every risk shows the measured evidence behind it; “right now” (hazards, heat, air quality, shelters, official contacts); methods, sources and limitations; QR code and share link. Three editions — **Citizen, Student, Planner** — in English or বাংলা. Save as PDF, download CSV, saved in the browser.

5.9 Ask Earth (AI guide)

A chat panel on every page. Answers come from Claude (when an Anthropic key is configured) or a free open AI model, grounded only in EarthScope's computed data and live feeds, with sources and follow-up questions. If AI is unavailable it falls back to a rule-based offline guide. It answers in the interface language (Bangla when বাংলা is selected).

6 The science: how every number is made

6.1 Detecting data breaks (homogeneity)

NASA POWER is a *reanalysis* (MERRA-2): a weather model constantly corrected by observations. When the set of observations changes (new satellites, stations lost), the series can jump. Over Bangladesh, POWER temperature drops about **1 °C in 1997** and rainfall jumps again around 2016 — not real climate.

Method: two-phase regression with a common slope (Lund & Reeves 2002). For each possible break year we fit "trend" vs "trend + step" and compute an F statistic; the maximum F is compared with 99% critical values we computed by Monte-Carlo simulation (20,000 random series per length). A break must also be large enough to matter (≥ 0.3 °C, or ≥ 15% of mean rainfall). Years before a break are shown grey and excluded from every trend. Code: `lib/homogeneity.ts`, `lib/climate.ts`.

6.2 Independent cross-check with ERA5

ERA5 (ECMWF / Copernicus) is a separate reanalysis with a different model and assimilation system. For each place EarthScope downloads ERA5 daily data (1981–2025), makes annual values and compares them with POWER: year-to-year correlation, average bias, trends over the same years, rainfall ratio. Most importantly it runs the break test on the **difference series POWER − ERA5**: a step there proves the break belongs to one dataset, not to the climate (the "reference series" method of relative homogenisation). Code: `lib/crosscheck.ts`, `/api/crosscheck`.

6.3 Honest trend statistics

- Trend = least-squares slope, reported per decade, with a **95% confidence interval**.
- Significance from a t-test **corrected for autocorrelation** (effective sample size, Santer et al. 2000) — climate years are not independent, and ignoring this makes trends look more certain than they are.
- The words "warming" or "cooling" appear only when $p < 0.05$; otherwise EarthScope says "no clear trend yet". Reports use IPCC calibrated likelihood language.
- Decade comparisons need ≥ 20 consistent years. Code: `lib/stats.ts`.

6.4 Climate extremes

From NASA POWER *daily* data, following the WMO ETCCDI index conventions with fixed thresholds: hot days (Tmax ≥ 35 °C), warm nights (Tmin ≥ 25 °C), humid-heat days (wet-bulb ≥ 27 °C), frost days, hottest day, heavy-rain days (≥ 50 mm), wettest day, longest dry spell. Computed only over the break-free years. Code: `lib/extremes.ts`.

6.5 Decoding satellite maps exactly

NASA GIBS serves measured products as coloured images using fixed published colour maps. EarthScope downloads the PNG images and converts each pixel back to its value using those colour maps:

- **Land cover** (MODIS MCD12Q1): 17 IGBP classes, each a unique colour → grouped into built-up, cropland, forest, shrub/grass, wetland, water, barren.
- **NDVI** (MOD13Q1): a 250-step palette → vegetation index.
- **Land-surface temperature** (MOD11A2): 250 colours in **0.6 K steps** from −73 °C to +77 °C. In testing, all 65,536 pixels of a Dhaka tile decoded exactly.

Comparisons always use the **same product and the same season** in both years, so a difference reflects the land, not the sensor. Code: `lib/earth-change.ts`.

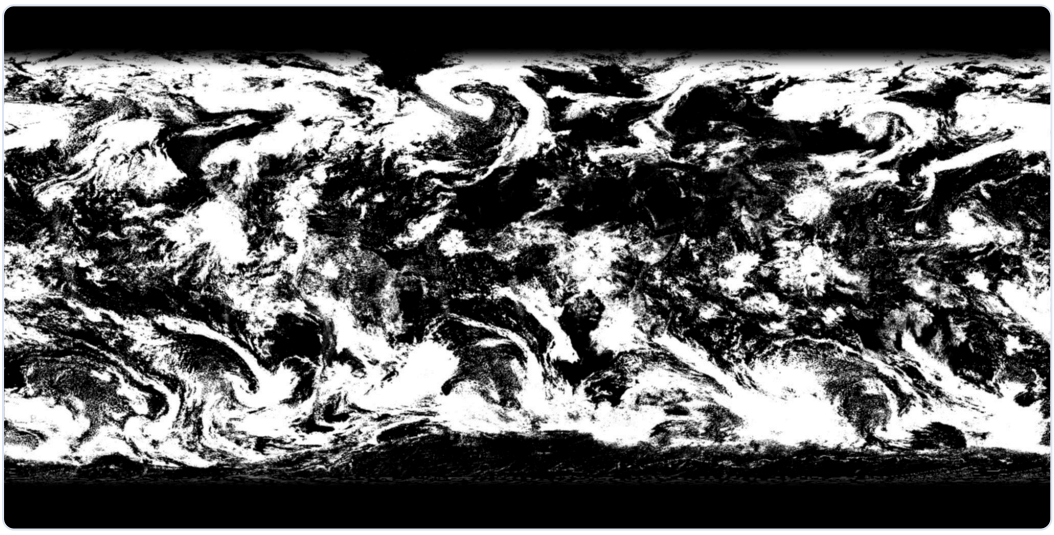
6.6 Surface heat: change relative to the surroundings

One April can be hotter than another everywhere because of weather. So besides the area-mean change, EarthScope removes the area-wide shift and marks pixels that warmed or cooled **≥ 1.5 °C more than the area as a whole** — typically new buildings and lost trees (warmer) or new water and parks (cooler). It also reports **hot-spot intensity** (mean of the hottest 10% minus the median), a standard surface urban-heat-island measure.

6.7 Air quality and sea level

- **Air quality**: Copernicus CAMS hourly forecast (PM2.5, PM10, ozone, NO_2) via Open-Meteo; US EPA AQI categories with official health advice; PM2.5 compared with the WHO 2021 guideline (15 µg/m³ daily). Code: `lib/air-quality.ts`.
- **Sea level**: IPCC AR6 WGI Table 9.9 global-mean projections (e.g. SSP2-4.5 in 2100: 0.56 m, likely 0.44–0.76 m) and observed rise since 1901 (≈0.20 m, now ≈3.7 mm/yr). Ground height from Copernicus GLO-90. Code: `lib/sea-level.ts`.

6.8 Daily real clouds on the globe



A real cloud mask built from NASA VIIRS imagery (white = cloud). Cyclones and weather fronts of that day are visible.

The server downloads the NOAA-20 and Suomi-NPP true-colour images of a day plus the previous 6 days. Clouds move but ice, snow and deserts don't, so the darkest value of each pixel over the past days is the ground; only what is brighter (and white) on the chosen day is cloud. Taking the darker of the two satellite passes removes ocean sun-glint. Built in about 8 seconds, cached for a year per date, pre-built every night by a scheduled job. Code: `lib/clouds.ts`, `/api/clouds`.

6.9 Hazard zones and safety

Impact zones are distance estimates from magnitude, wind speed or alert level (or measured footprints where GDACS provides them); cyclone positions along forecast tracks are interpolated in time. Shelters, hospitals and assembly points come from OpenStreetMap; any inside a hazard zone are hidden. This is a screening aid, never an official warning. Code: `lib/hazards.ts`, `lib/safe-places.ts`.

6.10 Trend Lab statistics (maps of many places)

- **Mann–Kendall test**: compares every pair of years and counts how often the later value is higher. Rank-based, so it is robust to outliers and to skewed variables such as rainfall. Includes the tie correction.
- **Trend-free pre-whitening** (Yue et al. 2002): remove the trend, measure lag-1 autocorrelation, remove it if significant, add the trend back, then test — so persistent year-to-year memory is not mistaken for a trend.
- **Sen's slope**: the median of all pairwise slopes — the standard robust "how much" paired with Mann–Kendall. Reported per decade.
- **False-discovery rate** (Benjamini–Hochberg, $\alpha = 10\%$, as recommended by Wilks 2016 for climate maps): with hundreds of cells, ~5% would look significant by chance at $p < 0.05$. FDR keeps the expected share of false "significant" cells below 10%.
- **Per-cell break test** for NASA POWER: the same Lund & Reeves test as for single places; if a step is found, only the years after it are tested (cells with fewer than 15 consistent years are excluded and shown grey).
- **Seasons**: monthly values are combined into a seasonal total (rainfall) or day-weighted mean; winter is assigned to the year of its January.
- **Satellite pixels**: MODIS monthly products decoded exactly from GIBS colour maps; land cover from MODIS MCD12Q1 (2001 and 2024) labels each pixel.

Code: `lib/trends.ts`, `lib/trend-lab.ts`, `components/trend-lab.tsx`, `/api/trend-grid`; tests in `tests/trends.test.ts` (including a textbook FDR example and a planted-step case).

7 Our key findings

7.1 Dhaka: the hidden warming

This is the strongest story to tell judges: it shows why careful data handling matters.

Question	Naive answer (raw NASA POWER)	EarthScope's answer
Is Dhaka warming?	Full record 1981–2025 shows **cooling** — because of the artificial 1997 drop.	The 1997 step is detected (≈ −1 °C) and excluded. POWER after 1997 shows **no clear trend** (0.00 °C/decade).
Is the 1997 drop real?	Cannot tell.	**No.** The POWER − ERA5 difference series itself jumps −1.0 °C in 1997–98: the fault is in POWER.
So what is the real trend?	—	ERA5 has no break and shows **+0.21 °C per decade** over 1981–2025 (+0.24 since 1997). Agreement POWER vs ERA5 after 1997 is weak (r = 0.47), so EarthScope shows the cross-check openly.
How has the land changed?	—	Built-up share of the 45 km Dhaka square: **13.9% (2001) → 19.5% (2024)**, about **109 km²** more. Outskirts warmed more than their surroundings (surface heat change map).
Is the air safe?	—	On test days the US AQI was 151–168 ("Unhealthy"), PM2.5 ≈ 6× the WHO daily guideline.
Sea level?	—	Central Dhaka ≈ 19 m above sea level → not directly flooded by the sea, but backwater, waterlogging and salinity reach inland; the delta is sinking several mm/year in many places.

> **How to say it.** "Raw NASA data would tell Dhaka it is cooling. EarthScope found a 1997 fault in the data, proved it with a second, independent dataset, and shows the real picture: about +0.2 °C per decade of warming, a city that grew by over 100 km² of concrete, and air that is unhealthy on many days."

Numbers were produced by the live app on 3 October 2026; air quality and hazards change daily.

7.2 Trend detective: how much of the map is real?

Trend Lab, NASA POWER, Bangladesh and neighbours (20–30°N, 80–100°E, 693 cells), 1981–2025, FDR 10%:

Variable · season	Naive analysis (no break test)	EarthScope (break-tested)
Air temperature · whole year	39% rising · **21% falling**	**36% rising · 8% falling** · 224 cells had a step (most often 2008)
Rainfall · monsoon (Jun–Sep)	69% rising · 0% falling	50% rising · 1% falling · 287 cells had a step

What this means. Without break testing, the share of the region that appears to be *cooling* is 2.6 times too large, and monsoon rainfall looks wetter almost everywhere. Even after break handling, the widespread POWER rainfall increase should be treated with caution (a second break or changing inputs can remain) — which is exactly why EarthScope shows the test, the breaks and the ERA5 cross-check instead of a single confident number.

The temperature map also shows opposite signs inside one region: warming over the Himalayan foothills, eastern Bangladesh and Myanmar, and a cooling pocket over parts of the Ganges plain — a pattern often linked in research to aerosol haze and irrigation (a regional "warming hole"). EarthScope flags it; confirming the cause would need further study.

7.3 One driver, opposite trends — irrigation around Dhaka

Trend Lab case study, NASA MODIS Terra, 240 km around Dhaka, every March 2001–2026, mean Sen's slope per decade by MODIS land cover; % = share of pixels with a significant trend (FDR 10%):

NASA variable (March)	Built-up land	Cropland	Reading
Daytime surface temperature	−0.03 °C/decade	**−0.58 °C/decade** (33% sig.)	Fields cool by day; city–field contrast grows +0.55 °C/decade
Night-time surface temperature	+0.41 °C/decade	**+0.41 °C/decade** (19% sig.)	The same fields warm at night — opposite to daytime
Vegetation greenness (NDVI)	−0.007/decade	**+0.047/decade** (79% sig.)	Fields green up (more irrigation); the city browns

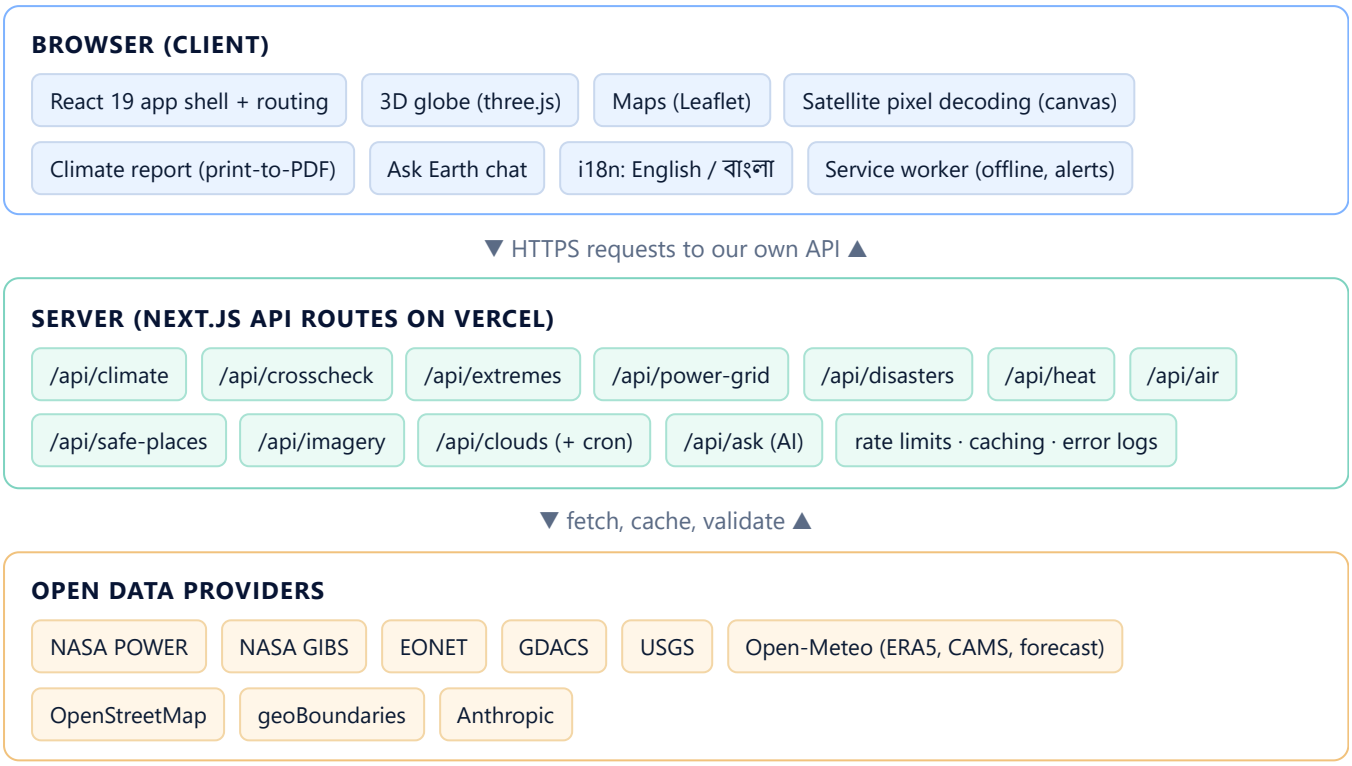
The mechanism. Dry-season boro rice irrigation has spread across the delta. Water evaporating from greener fields uses the Sun's energy, cooling the ground by day; at night there is no sunshine to drive evaporation, and wet soil and humid air hold heat, so the same fields warm. The city does not cool like its surroundings, so the surface urban heat island grows. Same process, opposite trends — by time of day and by land type. This matches published research on irrigated regions.

8 Data sources

Source	Used for	How often
NASA POWER (MERRA-2)	Monthly and daily temperature, rainfall, wet-bulb; atlas grid	On demand, cached 1 day
NASA GIBS	Blue Marble, Black Marble, VIIRS clouds, MODIS land cover, NDVI, LST, aerosols, true colour, HLS Landsat/Sentinel-2, GPM rain, VIIRS fires, LANCE floods, SEDAC population	Daily to yearly products
NASA EONET	Open natural events	Every 5 min
NASA GISTEMP v4, **NSIDC** sea ice, **NOAA GML** CO_2	Global context on the globe, in reports and Ask Earth	Built by script
ECMWF ERA5 (via Open-Meteo)	Independent cross-check of every record	Cached 30 days
Copernicus CAMS (via Open-Meteo)	Air-quality forecast	Cached 30 min
Open-Meteo forecast, geocoding, elevation	Heat outlook, place search, ground height (Copernicus GLO-90)	On demand
IPCC AR6 WGI	Sea-level projections, reference values	Static (published)
GDACS (UN / EC JRC)	Alert levels, cyclone tracks, footprints, surge	Every 5 min
USGS	Earthquakes M4.5+	Every 5 min
OpenStreetMap (Overpass)	Shelters, hospitals, assembly points, fire stations	On demand
geoBoundaries, **Natural Earth**	Country, state and district boundaries	Static / cached
Esri World Imagery	Close-zoom map imagery and the photo underlay	Map tiles
Anthropic Claude / free open model	Ask Earth answers (optional)	Per question, rate-limited

All sources are open or free for this use; attribution is shown in the app and in every report.

9 Architecture and technology



Why a server in the middle? It hides keys, combines and cleans feeds (e.g. EONET + GDACS + USGS into one hazard list), enforces rate limits, and lets the content-delivery network cache results at the right frequency so the app is fast and kind to the free data services.

Technology stack

Area	Technology
Framework	Next.js 16 (App Router) with React 19 and TypeScript; also builds for Cloudflare Workers via vinext/Vite
UI	Custom CSS (dark navy theme), Radix UI primitives, lucide icons, sonner toasts; self-hosted Noto Sans Bengali
3D & maps	three.js (custom day/night/cloud shader), Leaflet with NASA GIBS and Esri tiles
AI	Anthropic SDK (Claude, streaming, tool use) or any OpenAI-compatible free model; offline rule engine
Images	jpeg-js (server-side cloud mask), canvas decoding in the browser, sharp (link-preview image)
Quality	Vitest (51 tests), ESLint, TypeScript strict, i18n coverage checker
Hosting	Vercel (functions up to 60 s, CDN, daily cron); optional Upstash Redis for shared rate limits

Performance

- **Code splitting**: the app shell is under 48 KB; three.js (≈524 KB), Leaflet (≈148 KB) and the report (≈136 KB) load only when needed. Before: one 1.2 MB bundle.
- **Caching**: each API sets cache headers matched to how often its source changes (5 min for hazards, 30 min for air, 30 days for ERA5, 1 year per date for clouds).
- **Offline**: the service worker keeps the app shell and the last live data; the safety pack works without network.

10 Code map: where everything lives

```
app/                    Pages (one per view) and server API routes
  page.tsx, disasters/, atlas/, satellite/, future/, compare/, reports/
  api/*/route.ts        18 API routes (see section 9)
  layout.tsx            Site metadata, link preview, global CSS
  *.css                 Styles: terra, maps, earthscope (theme), report (print)
components/
  app/                  App shell and views
    shell.tsx           State, routing, language, service worker, update prompt
    nav.tsx, hero.tsx   Sidebar, page headings, animated planet header
    explore.tsx         Explorer page: metric cards, data-quality note, story
    crosscheck.tsx      ERA5 cross-check card
    views.tsx           Future, Compare, Reports views
    dialogs.tsx, lazy.tsx, error-boundary.tsx, charts.tsx, place-picker.tsx
  earth-globe.tsx       3D globe
  disaster-map.tsx      Live disasters + "Am I safe?"
  hazard-panels.tsx, my-places.tsx, air-quality.tsx
  climate-atlas.tsx     Atlas
  trend-lab.tsx         Trend Lab (grid + pixel engines, case study)
  satellite-compare.tsx Satellite comparison (4 modes, swipe, time-lapse)
  sea-level.tsx         Sea-level panel
  climate-report.tsx    8-page report
  ask-earth.tsx         AI chat
lib/                    Science and data code (no UI)
  stats.ts, homogeneity.ts, climate.ts, crosscheck.ts, extremes.ts
  trends.ts, trend-lab.ts  Mann–Kendall, Sen, FDR; grid and pixel trend analysis
  earth-change.ts       Land cover, NDVI, LST decoding and comparison
  air-quality.ts, sea-level.ts, heat.ts, hazards.ts, hazard-feed.ts
  safe-places.ts, safety-watch.ts, geo.ts, locate.ts, clouds.ts
  ask-ai.ts, ask-rules.ts  AI guide and offline guide
  i18n.tsx, i18n-bn.ts, climate-bn.ts   Languages
  rate-limit.ts, telemetry.ts           Protection and error reports
  report/               Report model, data gathering, translations, share links
tests/                  Vitest unit tests
scripts/                Data builders, i18n checker, OG image, build helpers
public/                 Textures, data files, icons, service worker, og.png
```

About 14,000 lines of project code (excluding generated UI primitives), 13 commits in the improvement programme.

11 Server API reference

Route	Purpose	Upstream
`/api/climate`	Monthly temperature and rainfall record 1981–2025 for a point	NASA POWER
`/api/crosscheck`	ERA5 annual series for the cross-check	Open-Meteo archive (ERA5)
`/api/trend-grid`	Monthly values of one NASA POWER variable for every cell of a 10° tile (Trend Lab)	NASA POWER regional
`/api/extremes`	Daily data → ETCCDI-style extreme indices	NASA POWER daily
`/api/power-grid`	10° tiles of gridded climate for the atlas	NASA POWER regional
`/api/boundaries`	State/district boundaries	geoBoundaries
`/api/disasters`	Unified live hazard feed with forecasts	EONET, GDACS, USGS
`/api/event-impact`	People exposed, storm surge for one event	GDACS, SEDAC
`/api/heat`	7-day heat index and BMD heatwave class	Open-Meteo forecast
`/api/air`	Air quality now + 5-day forecast	Copernicus CAMS
`/api/safe-places`	Shelters, hospitals, assembly points nearby	OpenStreetMap Overpass
`/api/elevation`	Ground height	Copernicus GLO-90
`/api/imagery`	Satellite snapshots (land cover, NDVI, LST, HLS, MODIS)	NASA GIBS WMS
`/api/clouds`, `/warm`	Daily cloud mask for the globe; nightly pre-build	NASA GIBS VIIRS
`/api/places`, `/api/locate`	Place search; approximate location fallback	Open-Meteo / GeoNames, IP lookup
`/api/ask`	Ask Earth, streamed answers	Claude or free model
`/api/log`	Receives browser error reports	—
`/api/version`	Current build id (open tabs offer “Reload”)	—

All routes validate their inputs (coordinates, dates, sizes) and costly routes are rate-limited per visitor with a daily cap per route.

12 Bangla (বাংলা) and accessibility

- **Whole interface in Bangla**: a switch at the top (EN | বাংলা), remembered per browser and chosen automatically for Bangla-language browsers.
- All numbers, years and dates in **Bangla digits** (২০২৫, ১৩.৯%); Bangla typography with Noto Sans Bengali.
- **How it works:** components call `t("English text")`; `lib/i18n-bn.ts` holds the Bangla translation keyed by the English text (830+ strings). Missing strings fall back to English, never break.
- **Safety net:** `npm run i18n:check` lists untranslated strings; the test suite fails if any is missing or if a translation drops a `{placeholder}`.
- The climate story, reports (all three editions) and Ask Earth answers are also in Bangla.
- Accessibility: keyboard-operable sliders (arrow keys, Home/End), screen-reader labels on controls, sufficient colour contrast, works on phones (tested at 375 px width).

> **Adding a new sentence?** Write it in English inside `t("…")`, run `npm run i18n:check -- --missing`, add the Bangla line to `lib/i18n-bn.ts`, run `npm test`.

13 Quality: tests, security, privacy, offline

Automated tests (51, all passing)

File	What it proves
`tests/stats.test.ts`	Trend slopes, confidence intervals, autocorrelation correction, t-table values, p-value formatting
`tests/homogeneity.test.ts`	Break test finds a planted step, ignores trends and tiny steps; **regression test on the real Dhaka record** (1997 break detected, no false "cooling")
`tests/crosscheck.test.ts`	Daily → annual aggregation, correlation, agreement verdict, break confirmed through the difference series
`tests/earth-change.test.ts`	Every LST colour decodes exactly; relative-warming logic; composite calendars
`tests/i18n.test.ts`	Translator, Bangla digits, every placeholder kept, 0 missing translations
`tests/trends.test.ts`	Normal distribution, Sen's slope robustness, Mann–Kendall detection and no false alarm in noise, Benjamini–Hochberg worked example (incl. p = 0), season aggregation, grid analysis with a planted 1997 step (and what happens without the break test)
`tests/data.test.ts`	AQI categories, IPCC sea-level consistency, extremes indices, share-link privacy and tamper safety

Security and privacy

- No accounts, no tracking. Saved places, reports and the safety pack stay in the user's browser.
- Your location is used only to run the check; share links round coordinates to ~1 km.

- Rate limits on every costly route (shared across servers with Upstash when configured); request size limits; input validation; markup stripped from shared links.
- AI keys stay on the server; the free-model mode warns users not to type personal details.
- Error boundaries keep one failing feature from blanking the app; browser errors are reported to `/api/log`.

14 Deployment and operations

- **Production:** earthscope-climate.vercel.app (Vercel project *earthscope-climate*; the old *terra-climate-time-machine* address still works).
- **Deploy:** `npx vercel deploy --prod --yes` from the project folder (requires being logged in to the team's Vercel account). Vercel runs `next build`.
- **Scheduled job:** `vercel.json` runs `/api/clouds/warm` daily at 00:15 UTC to pre-build the cloud map.
- **Updates reach users:** each build has an id; open tabs and installed apps check `/api/version` and show "A new version is available — Reload". The service worker uses a fresh cache per build.
- **Logs:** Vercel dashboard → project → Logs (search `client-error` for browser errors).

Optional settings (environment variables)

Variable	Effect
`ANTHROPIC_API_KEY`	Ask Earth uses Claude (best quality) and its tools
`UPSTASH_REDIS_REST_URL` / `_TOKEN`	Rate limits shared by all server instances
`CRON_SECRET`	Protects the nightly cloud job
`ASK_EARTH_RATE_LIMIT`, `ASK_EARTH_DAILY_LIMIT`	AI question limits (default 12 per 10 min, 500 per day)
`ASK_EARTH_OPENAI_URL` / `_KEY` / `_MODEL`	Use another free AI service (e.g. Groq)
`NEXT_PUBLIC_SITE_URL`	Public address used in link previews

Add them in Vercel → Settings → Environment Variables (never commit keys). None is required.

15 How to run, change and ship it

```
# one-time
npm install

npm run dev            # local app at http://localhost:5173
npm test               # 51 unit tests
npm run lint           # code style and React rules
npm run i18n:check     # Bangla coverage
npx next build         # production build (what Vercel runs)
npx vercel deploy --prod --yes   # publish
```

Team workflow

1. Pull the latest code; create a branch for your change.
2. Make the change; keep science code in `lib/` and add a test for it.
3. Any new visible text goes through `t()` with a Bangla translation.
4. Run `npm test`, `npm run lint`, `npx next build` — all must pass.

5. Check the page in the browser in both languages and on a phone-sized window.
6. Commit with a clear message; deploy; open the live site and verify.

Maintenance scripts

`node scripts/build-globe-textures.mts`	Refresh Blue Marble, Black Marble and the fallback cloud map
`node scripts/build-global-record.mts`	Refresh GISTEMP, sea ice and CO_2 global record
`node scripts/build-world-climate.mts`	Rebuild country climate series for the atlas
`node scripts/build-og-image.mjs`	Rebuild the link-preview image

16 Project history

Before the improvement programme

The project started as “Terra”, a climate time machine with the NASA POWER explorer, globe, atlas, live disasters with “Am I safe?”, satellite comparison, scenarios and reports. It was then renamed **EarthScope** with a new dark navy/blue design, a realistic three.js globe, an animated planet header, moving clouds, a rewritten satellite comparison and an 8-page bilingual climate report.

The 15-point improvement programme (3 October 2026)

#	Improvement	Result
1	Version control	Clean Git history; ready to publish on GitHub
2	README	Rewritten: features, trust, setup, settings, architecture
3	Link preview	1200×630 image from NASA textures + Open Graph/Twitter tags
4	Dead code	47 unused UI components, database scaffolding, 15 packages removed
5	Code splitting	Shell 1.2 MB → under 48 KB; heavy parts load on demand
6	Page URLs	Every view has its own address; back button works
7	Full Bangla	830+ strings, Bangla digits, Bangla AI answers, coverage test
8	Air quality	CAMS forecast card, NASA aerosol layer, safety status, report section
9	Sea level	IPCC AR6 projections + ground height + delta subsidence note
10	Surface heat	MODIS LST mode with exact decoding and relative-change map
11	ERA5 cross-check	Independent validation; confirmed the 1997 artefact
12	Tests	51 automated tests
13	Daily clouds	Server-built daily cloud map + nightly job
14	Protection	Rate limits (Upstash-ready) and error reporting
15	Refactor	1,000-line main file split into 9 focused modules

After the programme

- Land-cover map key, satellite-photo underlay and matching colours; “savanna” explained for Bangladesh.
- Land-cover **time-lapse** 2001–2024 with built-up growth chart.
- Update prompt for open tabs and installed apps.
- **Trend Lab** for the “Be An Earth System Trend Detective” challenge: grid and satellite-pixel trend maps with Mann–Kendall, Sen's slope, pre-whitening, FDR and per-cell break tests, plus the live opposite-trends case study. The new tests caught a real bug before release (a p-value of exactly 0 was treated as “nothing significant”).
- **Bug fixed:** the swipe showed the later year on both sides (a layering conflict introduced with the photo underlay). Found by reproducing on the live site, fixed and verified in all four modes.

17 Honest limitations

Know these before presenting — judges respect teams that state their limits.

Limitation	What it means / what we do about it
Reanalysis, not stations	NASA POWER and ERA5 are ~50 km / ~25 km grid estimates, not Dhaka weather-station readings. BMD station data would be the next step (not openly available).
POWER and ERA5 disagree after 1997 in Dhaka	Correlation is only 0.47; we show both and explain it rather than hide it.
Scenarios are not projections	The 2050 sliders are a simple sensitivity test, labelled as such; sea-level numbers are IPCC global means.
Hazard zones are estimates	Distance-based screening, not official warnings; shelters depend on OpenStreetMap volunteers' mapping.
Land-cover accuracy	MODIS land cover is ~74% accurate overall; changes under 1–2 points may be noise; Bangladesh rural land is often labelled “savanna”.
Satellite resolution	Land cover 500 m, LST 1 km, NDVI 250 m — good for city trends, not individual buildings (the 30 m view is visual only).
Surface vs air temperature	LST is the temperature of roofs and ground at ~10:30, hotter than air; clouds leave gaps.
POWER rainfall trends	Even after break handling, POWER rainfall rises almost everywhere around Bangladesh; treat it with caution and compare with ERA5 or GPM.
MODIS Terra orbit drift	Since about 2020 Terra's overpass time has been drifting earlier, which can slightly cool recent daytime surface temperatures. It affects city and fields alike, so the city–field contrast is robust; absolute daytime trends should be cross-checked with Aqua.
Field-significance choice	FDR at 10% is a standard choice for climate maps; a stricter level shows fewer significant cells. The map and the method are shown so experts can judge.
AI answers	Grounded and sourced, but can still be wrong; we say so in the chat.
Free services	Depends on free public APIs; caching, retries and fallbacks reduce but cannot remove outages.

18 Roadmap

Next weeks

- Publish the code on GitHub (open source)
- Add Upstash for shared rate limits
- Push notifications even when the app is closed
- Time-lapse for vegetation and surface heat

Next months

- BMD station data (partnership)
- FFWC river-level forecasts on the map
- CMIP6 downscaled projections for Bangladesh
- Ward-level heat-risk maps for Dhaka with population

19 Winning demo script and judge Q&A

5-minute winning demo

Time	Show	Say
0:00	Explorer, globe spinning	“NASA data say a lot about our planet — if you read them carefully. EarthScope is an Earth-system trend detective, built for Dhaka. These clouds are real, from NASA satellites two days ago.”
0:30	Dhaka cards → data-quality note → ERA5 card	“Raw NASA POWER says Dhaka is cooling. We detect a 1997 fault, prove it with an independent dataset, ERA5, and recover the real warming: about +0.2 °C per decade.”
1:15	Trend Lab → temperature map	“Now every grid cell in the region: red warming, blue cooling, dots significant after false-discovery-rate control. What, where, how much, and is it significant — one screen.”
1:50	Untick **Break test**, then tick it again	“Watch: a naive analysis says 21% of this region is cooling. With break testing it's 8%. Data artefacts, not climate. This is the trap most trend tools fall into.”
2:30	Run the case study live; Show the daytime map	“The challenge says one process can push trends in opposite directions. Here it is, from NASA MODIS pixels: irrigation cools Bangladesh's fields by day, warms the same fields at night, and they green up — while Dhaka's heat island grows.”
3:30	Satellite → Land cover → Time-lapse	“And why: 109 km² of new built-up land around Dhaka since 2001.”
4:00	Switch to বাংলা; Live disasters → Am I safe?	“All of it in Bangla, and useful today: hazards, heat, unhealthy air and the nearest shelters outside danger zones.”
4:30	Export a report (Planner, বাংলা)	“A report a city planner can print — with every method and source. EarthScope: honest trends, for the people who live them.”

Likely questions

Why Mann–Kendall and not just linear regression?	It is rank-based, so robust to outliers and non-normal data like rainfall; it is the standard test in climate trend studies. We still show regression with confidence intervals for single places.
What is the false-discovery rate and why use it?	Testing ~700 cells at $p < 0.05$ would mark ~35 as “significant” by pure chance. FDR (Benjamini–Hochberg, 10%) controls the share of false discoveries across the whole map — the approach recommended for climate maps (Wilks 2016).
How do you handle autocorrelation?	Trend-free pre-whitening before Mann–Kendall; for single places, an effective-sample-size correction (Santer et al. 2000).
How do you know the 1997 / 2008 steps are fake?	A statistical break test finds them, they appear in hundreds of cells at once, and the independent ERA5 dataset doesn't have them: the POWER − ERA5 difference jumps exactly then.

Is the irrigation story real or a coincidence?	Three independent NASA measurements agree with one physical mechanism (day cooling, night warming, greening on cropland), the result is computed live and is significant in a third to four-fifths of cropland pixels after FDR, and it matches published research on irrigated regions.
Why not use weather stations?	BMD station data are not openly available; we use two independent global datasets and state the limit.
Is this an official warning system?	No — it is educational. It always links to BMD, FFWC, DDM and 999/1090.
Can it work outside Bangladesh?	Yes — Trend Lab regions and pixel analyses work for any place; Bangladesh extras switch on automatically there.

20 Glossary

Anomaly	Difference from a reference period (e.g. +0.5 °C warmer than the first decade).
AQI	Air Quality Index (US EPA scale 0–500); above 150 is “Unhealthy”.
Autocorrelation	Neighbouring years resemble each other, so there is less independent information than the number of years suggests.
Break / inhomogeneity	An artificial jump in a data series caused by a change in how it was measured.
CAMS	Copernicus Atmosphere Monitoring Service (EU air-quality forecasts).
Confidence interval (95%)	The range that contains the true trend with 95% confidence.
ERA5	ECMWF's global reanalysis; used here to cross-check NASA POWER.
ETCCDI	WMO expert team that defines standard climate-extreme indices.
FDR (false-discovery rate)	Method that controls how many “significant” results on a map are false alarms when many places are tested at once.
GIBS	NASA Global Imagery Browse Services — serves satellite imagery and products.
Heat index	“Feels-like” temperature combining heat and humidity.
HLS	Harmonized Landsat Sentinel-2: 30 m imagery made comparable across satellites.
IGBP	The 17-class land-cover scheme used by MODIS.
IPCC AR6 / SSP	The UN climate panel's 6th assessment; SSPs are emission pathways (low to very high).
LST	Land-surface temperature: how hot the ground and roofs are.
Mann–Kendall test	Rank-based test for a consistent upward or downward trend.
MERRA-2	NASA's reanalysis behind NASA POWER.
MODIS / VIIRS	NASA satellite instruments imaging the whole Earth daily.
NDVI	Vegetation index from satellite light (higher = greener).
p-value	Chance of seeing a trend this strong if there were none; below 0.05 counts as significant.
Reanalysis	A weather model continuously corrected by observations, giving complete global data.
Sen's slope	Median of all pairwise slopes — a robust “how much per year”.
Surface urban heat island	Cities' ground surfaces being hotter than their surroundings.

EarthScope — Climate Time Machine · NASA Space Apps Challenge 2026 · earthscope-climate.vercel.app · This report describes the deployed version of 4 October 2026. EarthScope is an educational tool, not an official warning service: in Bangladesh follow BMD and DDM and call 999 in an emergency.